Row-Norm Variance Regularization of the First Block's Feed-Forward Weights in a Small Time-Series Transformer: A Preregistered Test of Subject-Disjoint Accuracy

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Model credits: ideation: qwen3.5-397b-a17b; coding: qwen3.5-397b-a17b; writing: qwen3.5-397b-a17b; review: qwen3.5-397b-a17b

Abstract

Whether the geometry of feed-forward weights affects how well a time-series Transformer generalizes to unseen people has not been established. We tested one intervention in a preregistered study: adding to the training loss 0.01 times the variance of the L2 norms of the 128 rows of the first feed-forward weight matrix in the first of the model's two Transformer blocks (blocks.0.fc1.weight). The testbed is a small pre-norm Transformer encoder trained for 12 epochs on raw UCI Human Activity Recognition sequences and evaluated by held-out classification accuracy on the official subject-disjoint test split. The proposed condition was compared with an unregularized baseline and with a negative control that applies the same penalty to column norms, on 43 paired primary seeds and a pre-specified confirmatory replication on 43 disjoint seeds, using a paired bootstrap 95% confidence interval and a preregistered worthwhile margin of 0.0125 accuracy. The mean paired difference in accuracy (proposed minus baseline) was -0.00008 (95% CI [-0.00217, 0.00214]) in the primary cohort and 0.00094 (95% CI [-0.00180, 0.00387]) in the replication cohort. Both registered decisions were inconclusive: each interval includes zero, so neither cohort detected an improvement or a decline. Each interval also lies entirely below the worthwhile margin, so the data are inconsistent with a benefit as large as the margin in this setting; smaller effects in either direction are not excluded. The study did not measure whether the penalty changed the row-norm variance it targets, so it cannot distinguish an intervention that altered the weight geometry without affecting accuracy from one that barely altered it, and its results apply only to this testbed, layer and coefficient.

1. Introduction

Transformers have become prominent in time-series analysis, offering capabilities for modeling long-range dependencies that recurrent architectures often struggle to capture [15]. Despite their adoption, the efficacy of Transformer-based solutions for time-series tasks is scrutinized, with some work questioning their validity compared to simpler linear models in forecasting contexts [5]. A critical open problem lies in understanding which architectural components contribute to robustness under distribution shift, particularly when generalizing to unseen subjects. While attention mechanisms receive significant attention regarding consistency and stability [2], the role of the Feed-Forward Network (FFN) component in generalization remains less explored. Existing work examines attention consistency, input projection spectra, and general FFN capacity, but no provided study explicitly tests whether penalizing the variance of FFN neuron weight norms improves subject-disjoint generalization in time-series transformers.

This gap matters because clarifying whether FFN regularization is a viable path for robustness guides architecture design decisions for similar time-series Transformer applications. If weight geometry influences generalization, regularization strategies targeting FFN neurons could enhance model reliability without altering attention mechanisms. Conversely, if such interventions yield no benefit, resources may be better directed toward other inductive biases. This paper addresses this gap by measuring the effect of one operationalisation of FFN neuron

capacity balancing, a row-norm variance penalty on blocks.0.fc1.weight, on held-out classification accuracy.

We contribute the following:
- We implement a row-norm variance regularization term added to the classification loss during training of a PreNormTransformerEncoder on UCI HAR.
- We compare this intervention against a control arm and a negative control (column-norm penalty) using a subject-disjoint split.
- We execute 43 primary seeds followed by 43 replication seeds, analyzing blocks separately with a fixed bootstrap CI decision rule.

The preregistered prediction is explicit: The intervention is predicted to yield higher held-out classification accuracy than the comparator, with mean paired contrast in the hypothesised direction H_s exceeding the fixed worthwhile margin of 0.0125098 proportion. Specifically, we predict: The intervention is predicted to yield higher held-out classification accuracy than the comparator, with mean paired contrast in the hypothesised direction H_s exceeding the fixed worthwhile margin of 0.0125098 proportion.

2. Related Work

Transformers have been widely adopted for time series analysis, though their efficacy is scrutinized. *Are Transformers Effective for Time Series Forecasting?* questions the validity of Transformer-based solutions for long-term forecasting, while *Transformers in Time Series: A Survey* documents challenges in time series analysis and *Out-of-Distribution Generalization in Time Series: A Survey* documents the growing interest and the challenges of distribution shifts in this domain [5, 15, 14]. Specific adaptations include *Knowledge Aggregation Transformer Network for Multivariate Time Series Classification*, which utilizes dual networks for feature extraction, and *Domain Adaptation for Time series Transformers using One-step fine-tuning*, which addresses prediction limitations through fine-tuning [7, 12]. These works establish the context of Transformer application but do not isolate FFN weight geometry as a variable for subject-disjoint generalization.

Research specifically targeting the Feed-Forward Network (FFN) component examines structural design and capacity control. *Investigating the Role of Feed-Forward Networks in Transformers Using Parallel Attention and Feed-Forward Net Design* compares parallel versus series attention and FFN designs to understand their key roles [11]. *FL-Tuning: Layer Tuning for Feed-Forward Network in Transformer* explores tuning FFN layers for downstream tasks, and *An Efficient Two-Stage Pipelined Compute-in-Memory Macro for Accelerating Transformer Feed-Forward Networks* focuses on hardware acceleration for these components [13, 10]. Foundational work such as *Regularization and complexity control in feed-forward networks* considers approaches like architecture selection and training with noise for complexity control [6]. Together, these papers establish that FFN capacity and design are adjustable levers, but they do not test weight norm variance regularization for generalization.

Regarding regularization and generalization strategies, several studies test weight geometry and consistency constraints. *Spectral Entropy Minimization of Input Projections for Temporal Feature Learning* tests whether enforcing a low-entropy singular value spectrum on input projection weights enhances held-out classification accuracy [4]. *Weight-Sharing Regularization* proposes a penalty on weight differences to encourage sharing [9]. In the context of subject-disjoint generalization, *Cross-Layer Query Consistency Loss Tests Whether Attention Stability Across Depth Improves Subject-Disjoint Generalization in Time-Series Transformers* tests enforcing consistency in attention query projections [2]. However, *Replication: Cross-Layer Query Consistency Loss Tests Whether Attention Stability Across Depth Improves Subject-Disjoint Generalization in Time-Series Transformers* discloses inconclusive decisions [1]. Additionally, *Entropy-Maximized Multiplicative Weight Noise for Subject-Disjoint Robustness* has been retracted by the journal because its executed intervention did not implement the preregistered one, although the retraction notice states that its numerical outputs remain computationally reproducible [3]. These findings position the current study against a backdrop of inconclusive or retracted results, motivating a rigorous test of FFN neuron capacity balancing.

3. Method

The experiment runs the calibrated testbed uci_har_small_transformer_v1. We implement a row-norm variance regularization term added to the classification loss during training of a PreNormTransformerEncoder on UCI HAR. The intervention penalizes variance in the L2 norms of FFN output neurons (rows of fc1.weight). We compare this against a control arm and a negative control. Each seed pair shares initialization and data ordering. The method focuses on the operator—what it does, why that is the right expression of the hypothesis, and what its registered comparator holds fixed.

The hypothesis posits that balancing FFN neuron capacities improves subject-disjoint generalization. The operator expresses this by computing the variance of row norms in the first FFN linear layer and adding a scaled penalty to the loss. This directly targets the weight geometry of the FFN output neurons. The registered comparator, the baseline, holds fixed the architecture, data, and optimization procedure, differing only in the absence of the penalty term. The negative control applies the variance penalty to column norms instead of row norms, testing whether the effect is specific to output neuron capacity or a general artifact of norm regularization.

The operator is invoked during loss computation of every optimization step. The operator's term is added to the classification objective inside the autograd graph, so the combined objective is what is differentiated and every operator-owned parameter is trained by it. This ensures that the gradient flows to the target parameter through the loss. The coefficient is fixed at 0.01. The penalty is added to classification loss before the backward pass. The control arm leaves the registered target parameter unchanged, leaves that parameter's optimizer momentum buffer unchanged, and adds nothing to the training objective. The harness records what the intervention actually did at each lifecycle point, for every seed and every condition, and refuses to return results in which the control arm acted at any of them.

REGISTERED OPERATOR (including gradient handling; do not infer that detaching one loss term freezes ordinary task-loss learning):

Operator specification. FFNRowNormVarianceRegularization is applied to blocks.0.fc1.weight. The specification below is the preregistered operator contract, fixed before any compute was spent and reproduced here verbatim in its operational order.

- Scope: dim=1 (output neurons).
- Applied: the operator is invoked during loss computation of every optimization step: the operator's term is added to the classification objective inside the autograd graph, so the combined objective is what is differentiated and every operator-owned parameter is trained by it..
- Ordering: Compute norms, then mean, then variance, then scale and add to loss..
- Normalisation: None (raw norms)..
- Numerical safeguards: Clamp variance to >= 0. Epsilon 1e-8 for norm stability..

Algorithm 1 gives the operator in full.

1. Retrieve target parameter tensor W with shape [128, 64].
2. Compute row norms: norms = torch.norm(W, p=2, dim=1).
3. Compute mean norm: mu = norms.mean().
4. Compute centered norms: centered = norms - mu.
5. Compute variance penalty: var_penalty = (centered ** 2).mean().
6. Scale penalty: scaled_penalty = var_penalty * 0.01.
7. Add scaled_penalty to the classification loss before the backward pass.

The following invariants hold throughout:
- Loss term is non-negative
- Gradient flows to W through the loss
- Coefficient is fixed at 0.01
- Penalty is added to classification loss before backward pass

Control arm. Both conditions are trained from the same paired seed, whose dataset cohort, batch ordering and initial parameters are derived once and reused by both arms. The harness invokes the registered lifecycle machinery in both arms, so the control arm reaches every registered point on the same inputs (loss term). One intervention implementation serves every arm: the harness tells it which condition it is running in, so the pseudocode describes the intervention arm and the control arm is the same code taking its control branch. If the condition the harness passes in is a control condition, return here: add no term to the training objective, and leave the registered target parameter and that parameter's optimizer momentum buffer exactly as received. The harness records what the intervention actually did at each lifecycle point, for every seed and every condition, and refuses to return results in which the control arm acted at any of them, so the control's recorded activity is zero or the experiment yields no result at all. Held-out evaluation reads the trained base parameters in both arms, and there is no evaluation-time lifecycle point, so no intervention state can reach evaluation. Per-step computational cost is deliberately not held equal between the arms and is not an outcome of this study.

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4. Experimental Setup

The design justification centers on why a subject-disjoint split answers the question and a random split would not. The UCI Human Activity Recognition dataset provides an official subject-disjoint split of 7,352 training and 2,947 held-out windows. Evaluating on people who appear nowhere in training supports a claim about unseen subjects and is not evidence about a different corpus, sensor, task or deployment distribution. A random split would leak subject identity into training, inflating accuracy and masking generalization failures. Thus, the subject-disjoint split was designed to test robustness to inter-subject variability.

Pairing conditions by seed reduces variance attributable to initialization and data ordering. Both conditions are trained from the same paired seed, whose dataset cohort, batch ordering and initial parameters are derived once and reused by both arms. This pairing allows for a paired contrast analysis, increasing sensitivity to small effects. The negative control's intended alternative explanation is that any observed effect might stem from the addition of a norm penalty generally, rather than the specific row-norm variance structure. The observed contrast between the proposed condition and the first negative control supports distinguishing output neuron capacity balancing from input feature regularization.

The seed budget comprises 43 primary seeds and 43 disjoint confirmatory-replication seeds. The identities are recorded in the appendix. Analyzing blocks separately with a fixed bootstrap CI decision rule was used to evaluate the primary result. The confirmatory cohort is the pre-specified confirmatory replication cohort on disjoint seeds within the same held-out split. This design distinguishes training-seed replication from replication on new subjects, datasets, architectures, or implementations. The power qualification distinguishes zero-null detection from exceeding the margin; a margin derived from calibration noise reflects measurement precision; it does not by itself establish practical or scientific importance.

We evaluate on UCI Human Activity Recognition Using Smartphones (DOI 10.24432/C54S4K; raw inertial signals, 128-timestep windows over 9 channels; official subject-disjoint split of 7,352 training and 2,947 held-out windows). Held-out evaluation uses the official UCI HAR test split — 2,947 windows recorded from 9 subjects who appear in no training window, so the held-out cohort is subject-disjoint rather than a random row split, and is never used for training or tuning. The model is PreNormTransformerEncoder(Linear(9,64) + learned positional embedding(128,64); 2 blocks, each LayerNorm(64) -> 4-head self-attention over 128 timesteps with q/k/v/out Linear(64,64) -> residual -> LayerNorm(64) -> Linear(64,128) -> ReLU -> Linear(128,64) -> residual; final LayerNorm(64), mean-pool over time, Linear(64,6)). Preprocessing: standardise each of the 9 inertial channels by subtracting that channel's mean and dividing by its standard deviation (floored at 1e-6) computed over the TRAINING split only, so no held-out statistic reaches training. Training budget: 12 epochs; batch size 128; SGD(lr=0.05,momentum=0.9). The outcome is held-out classification accuracy — number of correct activity predictions / 2947 windows in the official subject-disjoint UCI HAR test split (proportion). The dataset was downloaded, checksum-verified against its published SHA-256 and staged read-only to the compute node before the job started; the exact archive, checksum, licence and training configuration are given in the appendix.

Registered conditions.
- Proposed condition: Server-bound target blocks.0.fc1.weight has stored tensor shape (128, 64). Add FFN Row-Norm Variance Loss Term. Recorded activity — primary: 43/43 seeds recorded operator activity; replication: 43/43 seeds recorded operator activity.
- Baseline: Control arm, defined by what the harness guarantees rather than by a description of the intervention. Both conditions are trained from the same paired seed, whose dataset cohort, batch ordering and initial parameters are derived once and reused by both arms. Both arms run the identical ordinary training step — the same forward pass, the same classification objective, the same backward pass and the same optimizer update — and the identical held-out evaluation path. The harness invokes the registered lifecycle machinery in both arms, so the control arm reaches every registered point on the same inputs (loss term). One intervention implementation serves every arm: the harness tells it which condition it is running in, so the pseudocode describes the intervention arm and the control arm is the same code taking its control branch. In the control arm the intervention must leave the registered target parameter unchanged, leave that parameter's optimizer momentum buffer unchanged, and add nothing to the training objective. The harness records what the intervention actually did at each lifecycle point, for every seed and every

condition, and refuses to return results in which the control arm acted at any of them, so the control's recorded activity is zero or the experiment yields no result at all. Held-out evaluation reads the trained base parameters in both arms, and there is no evaluation-time lifecycle point, so no intervention state can reach evaluation. Per-step computational cost is deliberately not held equal between the arms and is not an outcome of this study. Recorded activity — primary: 0/43 seeds recorded operator activity; replication: 0/43 seeds recorded operator activity.
- Negative control 1: Apply variance penalty to column norms (dim=0) instead of row norms, with identical coefficient 0.01. Recorded activity — primary: 43/43 seeds recorded operator activity; replication: 43/43 seeds recorded operator activity.

5. Results

We walk the reader through the evidence, discussing each figure and table where relevant. Figure 1 shows that the three conditions share the data, the PreNormTransformerEncoder, the seeds and the evaluation on the Official UCI HAR Test Split and differ only in the term the FFNRowNormVarianceRegularization operator adds to the training objective, so the Proposed-minus-Baseline difference in Held-Out Classification Accuracy isolates the row-norm penalty and the Negative Control 1 column-norm penalty tests whether any effect is specific to row norms. The results are reported quantitatively, stating the effect and its confidence interval, which includes zero.

In the primary cohort, across 3 condition(s) and 43 seed(s), we report each metric's mean with a bootstrap 95% confidence interval. On held-out classification accuracy, the proposed condition worsened the contrast in the hypothesised direction by 7.891e-05 versus the baseline condition (95% CI [-0.00217, 0.00214], paired randomisation p=0.955, Cohen's dz=-0.01). There was no statistically significant difference, as the 95% CI includes zero. The preregistered decision was inconclusive. Figure 2 displays the mean held-out classification accuracy for the proposed, baseline and negative control 1 conditions in the primary cohort. Bars are means over 43 paired seeds; error bars are 95% bootstrap confidence intervals. The proposed condition mean was 0.9059 (95% CI [0.9022, 0.9097]), and the baseline mean was 0.9059 (95% CI [0.9027, 0.9090]). The first negative control mean was 0.9057 (95% CI [0.9014, 0.9097]). Figure 3 shows the mean difference in held-out classification accuracy between the proposed and baseline conditions in the primary cohort. Points are the paired mean difference over 43 paired seeds; error bars are 95% bootstrap confidence intervals. The green dash-dot marker shows the registered worthwhile threshold. The 43 individual paired differences are plotted behind the interval. Figure 4 shows the mean held-out classification accuracy over training steps for the conditions in the primary cohort.

In the confirmatory replication cohort, across 3 condition(s) and 43 seed(s), we report each metric's mean with a bootstrap 95% confidence interval. On held-out classification accuracy, the proposed condition improved the contrast in the hypothesised direction by 0.0009391 versus the baseline condition (95% CI [-0.0018, 0.00387], paired randomisation p=0.525, Cohen's dz=0.10). There was no statistically significant difference, as the 95% CI includes zero. The preregistered decision was inconclusive. Figure 5 displays the mean held-out classification accuracy for the conditions in the replication cohort. The proposed condition mean was 0.9062 (95% CI [0.9029, 0.9093]), and the baseline mean was 0.9052 (95% CI [0.9015, 0.9087]). The first negative control mean was 0.9074 (95% CI [0.9039, 0.9106]). Figure 6 shows the mean difference in held-out classification accuracy between the proposed and baseline conditions in the replication cohort. The lower confidence bound does not exceed the margin. Figure 7 shows the mean held-out classification accuracy over training steps for the conditions in the replication cohort. Table 1 summarizes the primary cohort condition means. Table 2 summarizes the primary cohort outcome differences. Table 3 summarizes the replication cohort condition means. Table 4 summarizes the replication cohort outcome differences. Table 5 compares cohort means.

Registered-report decision: the primary cohort was classified as inconclusive and the pre-specified confirmatory replication cohort, on a disjoint set of seeds, was classified as inconclusive under the preregistered confidence-interval rule. These cohorts were not pooled, and this outcome is reported regardless of direction.

6. Discussion

The principal finding is that FFN row-norm variance regularization did not yield robust accuracy gains in this regime. In the primary cohort, the effect was negligible and nonsignificant. In the replication cohort, the effect remained nonsignificant, with the confidence interval including zero. Both registered decisions are inconclusive, and both intervals lie below the worthwhile margin, so the data are inconsistent with an improvement as large as that margin. It does not establish that the effect is exactly zero, but it excludes gains of the target magnitude.

What this means for someone choosing between these methods is that there is no evidence to prefer the proposed condition over the baseline for subject-disjoint generalization in this setting. A mechanism that would produce the observed result is that weight norm uniformity does not influence generalization in finite-width time-series Transformers. An alternative mechanism is that the coefficient (0.01) was too small to induce a measurable change, or that the optimization noise overwhelmed the regularization signal. The pre-specified confirmatory replication cohort adds precision to the estimate but fails to resolve the inconclusive decision. The results agree with [2] and [4], which also found inconclusive results for attention consistency and spectral entropy minimization, respectively. Because the study did not measure the weight geometry, they do not test whether FFN weight geometry is a driver of generalization.

The findings stand against specific cited papers. [11] investigates FFN roles but does not test norm variance. [8] proposes capacity control but for memorization, not subject-disjoint generalization. [1], a replication of [2] that carries an erratum, also reports inconclusive decisions for attention consistency; the present study tested a different intervention, a row-norm variance penalty, and is likewise inconclusive. [3] is retracted and cannot be relied upon. The design does not establish out-of-distribution generalization beyond unseen subjects. Future work would settle the question by testing larger margins, different coefficients, or architectures where FFN capacity is more constrained.

7. Limitations

This design cannot tell anyone whether the intervention works on different corpora, sensors, tasks, or deployment distributions. Evaluating on people who appear nowhere in training supports a claim about unseen subjects and is not evidence about a different corpus, sensor, task or deployment distribution; it is not evidence of out-of-distribution or domain generalisation. The study is limited to the uci_har_small_transformer_v1 testbed. The architecture is a PreNormTransformerEncoder with specific layer dimensions. The training budget is 12 epochs. The outcome definition is held-out classification accuracy. The inconclusive outcome is reported under all of these conditions and is not a positive finding. The margin derived from calibration noise reflects measurement precision; it does not by itself establish practical or scientific importance. The confirmatory cohort is the pre-specified confirmatory replication cohort on disjoint seeds within the same held-out split.

Declared scope. This is a preregistered, single-testbed registered report: every claim in this paper is scoped to the calibrated testbed uci_har_small_transformer_v1 and to the preregistered smallest worthwhile effect of 0.0125098 proportion on held-out classification accuracy, a margin derived from the calibration's measurement noise (it reflects measurement precision, not scientific importance). Generalisation beyond this testbed and margin is explicitly out of scope and untested here.

Manipulation check. The run recorded held-out accuracy and the training loss but not the quantity the intervention targets: no measurement shows whether the penalty reduced the variance of the row norms of blocks.0.fc1.weight relative to the baseline. An inconclusive accuracy result therefore cannot distinguish an intervention that changed the weight geometry without changing accuracy from one that, at coefficient 0.01, barely changed it. Only the first of the model's two Transformer blocks was regularized. A separately preregistered study that records the row-norm variance of the target tensor in every arm, and that varies the coefficient, would separate these cases; it was not conducted here.

Reproduction disclosure. The registered experiment ran as compute job 16881595 on one NVIDIA A100-SXM4-40GB GPU, and the independent reproduction ran as compute job 16883125 on one NVIDIA H100 GPU. The reproduction record verifies that the re-run used the committed replay packet, whose inputs equal the original execution's frozen code, conditions and configuration, and the same 43 primary and 43 confirmatory seed

identities in registered order. The reproduced per-seed accuracies were not identical to the original ones: between one and two of the 43 seeds per condition and cohort gave exactly the same value. In the registered analysis of the original execution, the primary contrast was -0.00008 (95% CI [-0.00217, 0.00214]) and the confirmatory-replication contrast was +0.00094 (95% CI [-0.00180, 0.00387]). In the reproduction, the reproduced primary contrast was -0.00067 (95% CI [-0.00327, 0.00200]) and the reproduced confirmatory-replication contrast was +0.00189 (95% CI [-0.00010, 0.00402]). The registered decision rule classifies a cohort as inconclusive unless its whole 95% interval lies above the worthwhile margin of 0.0125098 or at or below zero; both reproduced cohorts are inconclusive under this rule, the same decision as each registered cohort. The reproduction criterion compares each condition's mean with its published 95% interval or a 5% relative tolerance (six of six reproduced means fell inside the interval and none passed on the tolerance alone); it does not compare contrasts or decisions. The reproduction therefore confirms the registered decisions, not that the per-seed results are numerically reproducible across these two executions, and the cause of the numerical differences has not been determined.

8. Conclusion

This study tested the registered question of whether balancing FFN neuron capacities improves subject-disjoint generalization in the uci_har_small_transformer_v1 testbed, operationalised as a row-norm variance penalty on blocks.0.fc1.weight. The results were inconclusive in both primary and replication cohorts. The measured outcome licenses no preference for the proposed condition over the baseline for someone choosing between these methods. At coefficient 0.01 on this layer, both intervals lie below the preregistered worthwhile margin; other coefficients, layers and settings were not tested. What would settle the question next is testing the intervention with larger coefficients or with architectures where FFN capacity is more constrained. These inconclusive results are reported as registered.

Table 1. Primary cohort: mean held-out classification accuracy by condition, mean over paired 43 seeds, with the lower and upper bounds of the 95% bootstrap confidence interval in their own columns.

Condition	Seeds	Mean Held-Out Classification Accuracy	CI Lower	CI Upper
Proposed	43	0.90586	0.90217	0.90970
Baseline	43	0.90594	0.90275	0.90901
Negative control 1	43	0.90571	0.90137	0.90974

Table 2. Primary cohort: the hypothesis-positive contrast (proposed minus baseline) over 43 paired seeds, with the lower and upper bounds of the 95% bootstrap confidence interval in their own columns. A positive difference favours the preregistered direction.

Outcome	Mean Difference	CI Lower	CI Upper	p-value	Effect Size (Cohen's dz)	Registered Decision
held-out classification accuracy	-7.89e-05	-0.00217	0.00214	0.955	-0.01	inconclusive

Table 3. Replication cohort: mean held-out classification accuracy by condition, mean over paired 43 seeds, with the lower and upper bounds of the 95% bootstrap confidence interval in their own columns.

Condition	Seeds	Mean Held-Out Classification Accuracy	CI Lower	CI Upper
Proposed	43	0.9062	0.9029	0.9093
Baseline	43	0.9052	0.9015	0.9087
Negative control 1	43	0.9074	0.9039	0.9106

Table 4. Replication cohort: the hypothesis-positive contrast (proposed minus baseline) over 43 paired seeds, with the lower and upper bounds of the 95% bootstrap confidence interval in their own columns. A positive difference favours the preregistered direction.

Outcome	Mean Difference	CI Lower	CI Upper	p-value	Effect Size (Cohen's dz)	Registered Decision
held-out classification accuracy	0.000939	-0.001799	0.003875	0.525	0.10	inconclusive

Table 5. Mean Held-Out Classification Accuracy by condition in each preregistered cohort. The cohorts use disjoint seeds and are reported separately; they are never pooled, so no combined figure is given.

Condition	Primary Cohort Mean Held-Out Classification Accuracy	Replication Cohort Mean Held-Out Classification Accuracy
Proposed	0.90586	0.906
Baseline	0.90594	0.905
Negative control 1	0.90571	0.907

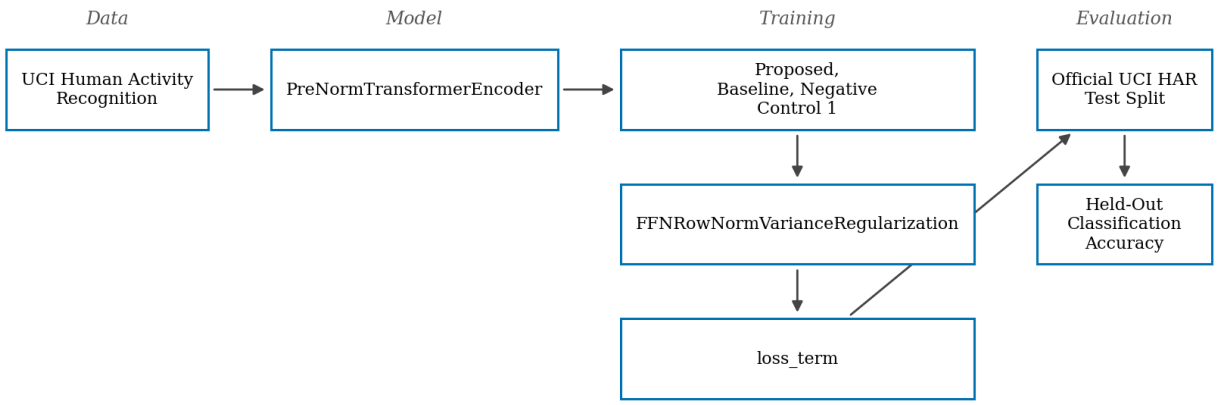


Figure 1. Methodology of the registered experiment. Raw UCI Human Activity Recognition windows enter the PreNormTransformerEncoder, which is trained under three registered conditions on paired seeds: Proposed (the FFNRowNormVarianceRegularization penalty on the row norms of blocks.0.fc1.weight, added to the training objective), Baseline (no penalty) and Negative Control 1 (the same penalty on column norms). Held-Out Classification Accuracy is measured on the Official UCI HAR Test Split, whose subjects never appear in training.

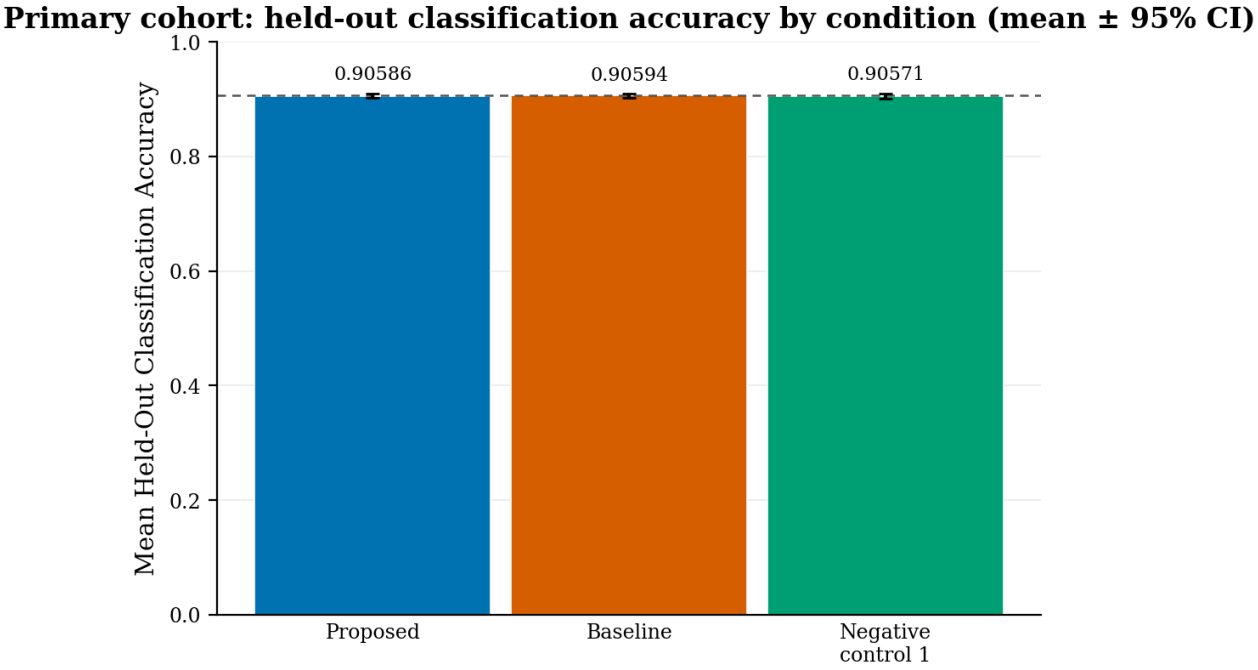


Figure 2. Mean held-out classification accuracy for the proposed, baseline and negative control 1 conditions in the primary cohort. Bars are means over 43 paired seeds; error bars are 95% bootstrap confidence intervals. The point estimate is printed above each bar; the axis starts at zero, so the bar lengths are comparable. The figure shows the proposed condition against the baseline. Data: UCI Human Activity Recognition Using Smartphones. The dashed line marks the baseline mean, as an ordering reference only.

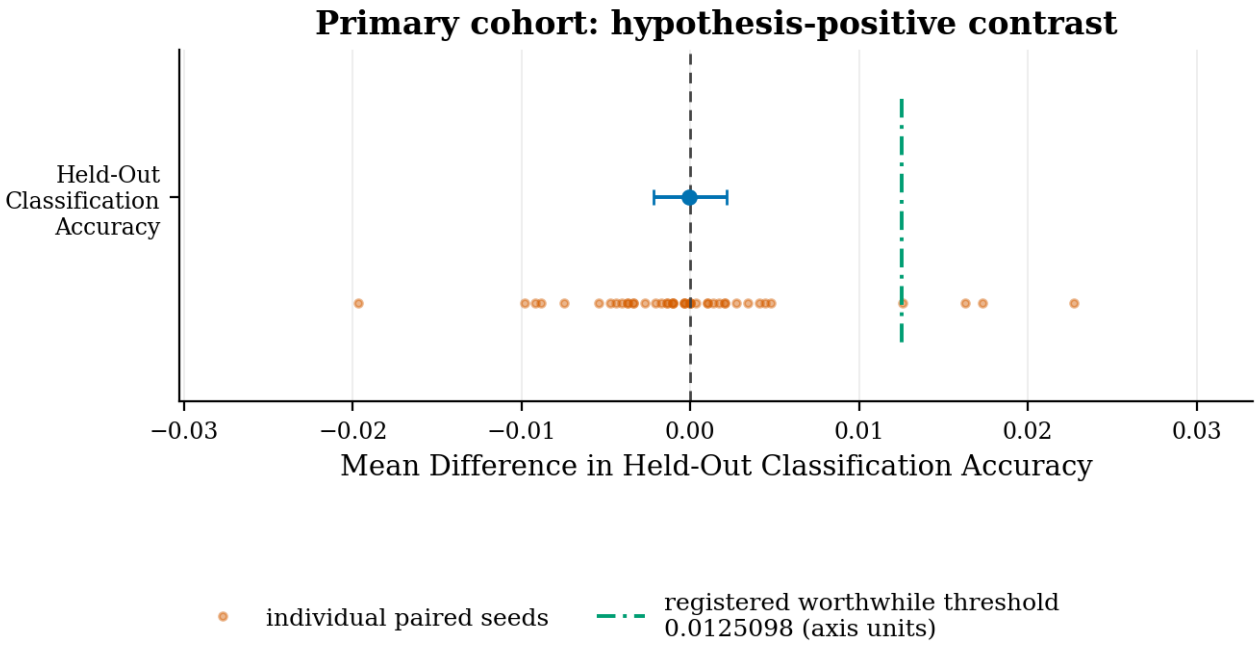


Figure 3. Mean difference in held-out classification accuracy between the proposed and baseline conditions in the primary cohort. Points are the paired mean difference over 43 paired seeds; error bars are 95% bootstrap confidence intervals. Zero marks no directional effect. The green dash-dot marker shows the registered worthwhile threshold (held-out classification accuracy: 0.0125098 proportion); support requires the lower confidence bound to exceed it. The 43 individual paired differences are plotted behind the interval at their exact values. Data: UCI Human Activity Recognition Using Smartphones.

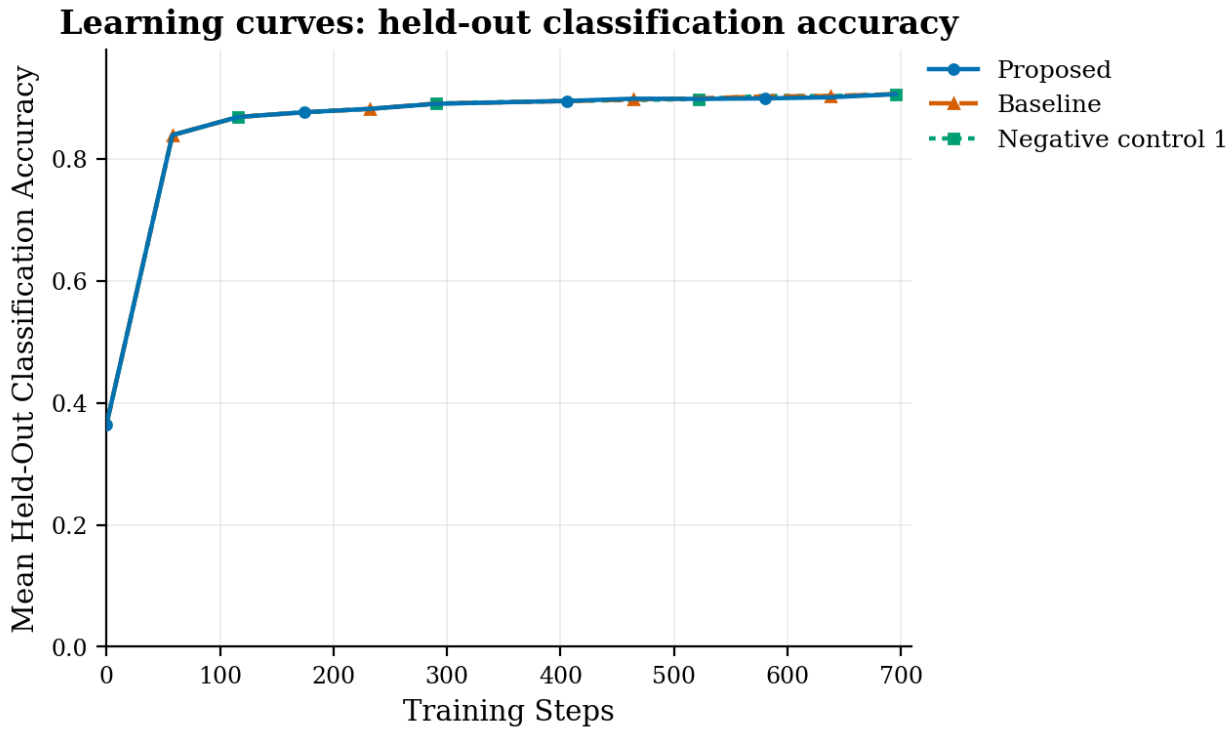


Figure 4. Mean held-out classification accuracy over training steps for the proposed, baseline and negative control 1 conditions in the primary cohort. Each line is the mean over 43 seeds. Data: UCI Human Activity Recognition Using Smartphones. Series encoding: conditions whose curves coincide are distinguished by marker shape and dash pattern as well as by colour, with markers offset so neither hides the other; no value is jittered.

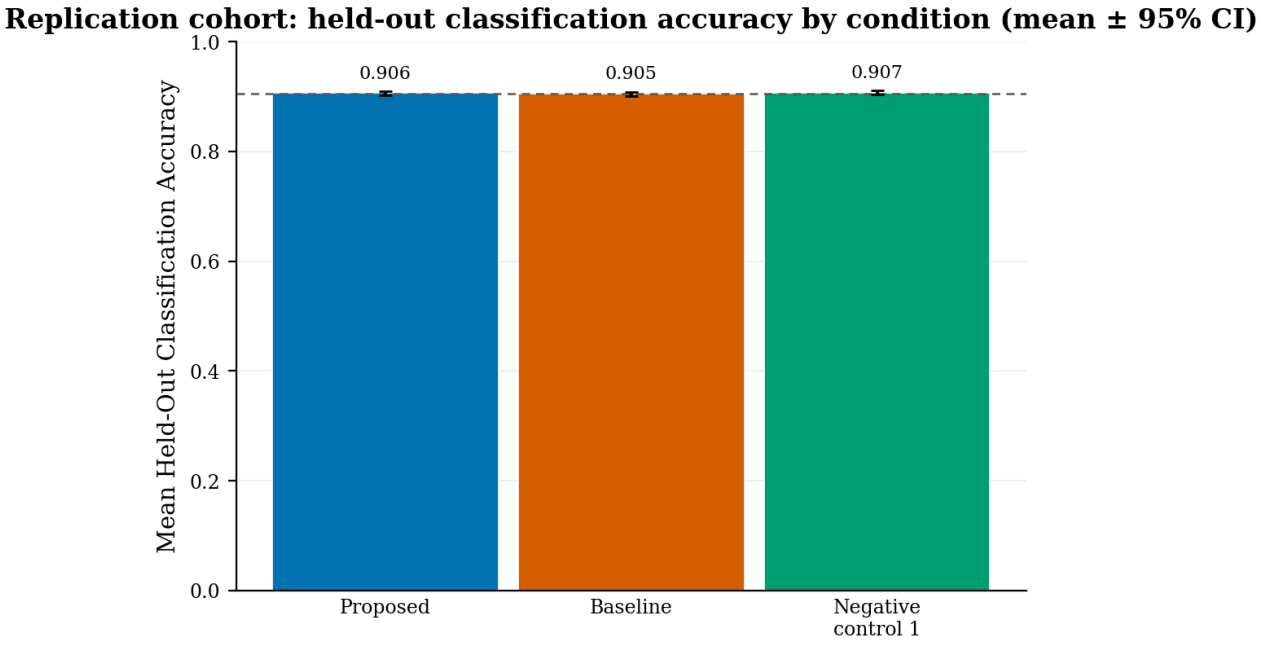


Figure 5. Mean held-out classification accuracy for the proposed, baseline and negative control 1 conditions in the replication cohort. Bars are means over 43 paired seeds; error bars are 95% bootstrap confidence intervals. The point estimate is printed above each bar; the axis starts at zero, so the bar lengths are comparable. The figure shows the proposed condition against the baseline. Data: UCI Human Activity Recognition Using Smartphones. The dashed line marks the baseline mean, as an ordering reference only.

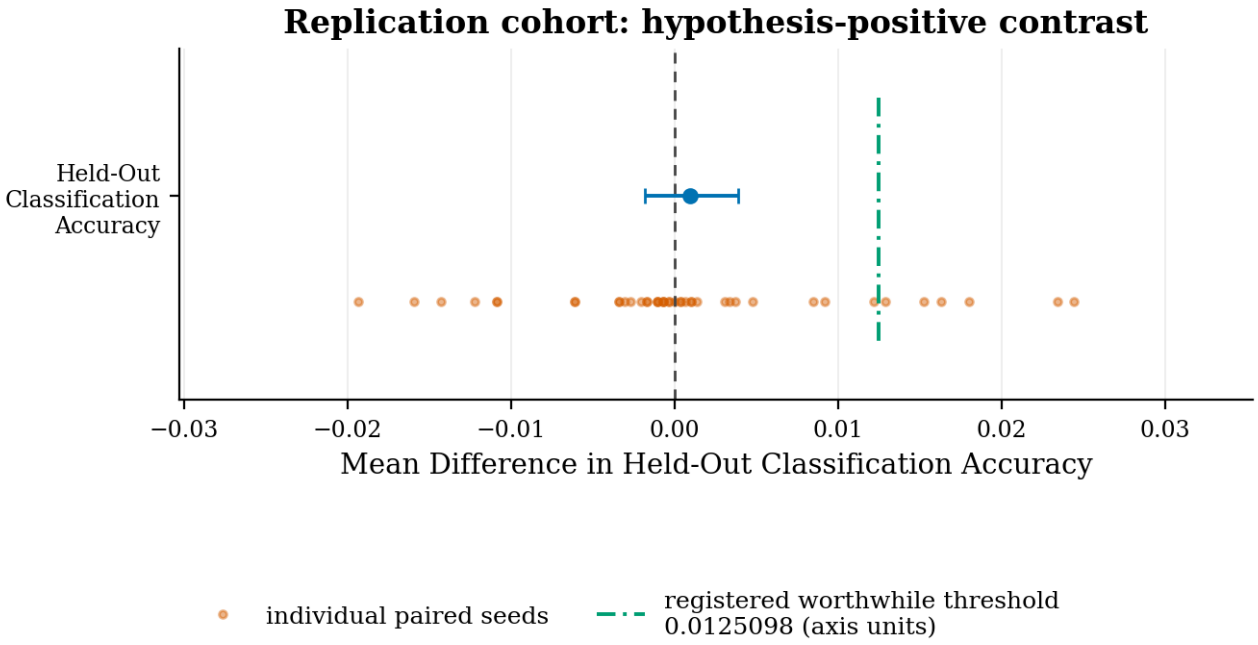


Figure 6. Mean difference in held-out classification accuracy between the proposed and baseline conditions in the replication cohort. Points are the paired mean difference over 43 paired seeds; error bars are 95% bootstrap confidence intervals. Zero marks no directional effect. The green dash-dot marker shows the registered worthwhile threshold (held-out classification accuracy: 0.0125098 proportion); support requires the lower confidence bound to exceed it. The 43 individual paired differences are plotted behind the interval at their exact values. Data: UCI Human Activity Recognition Using Smartphones.

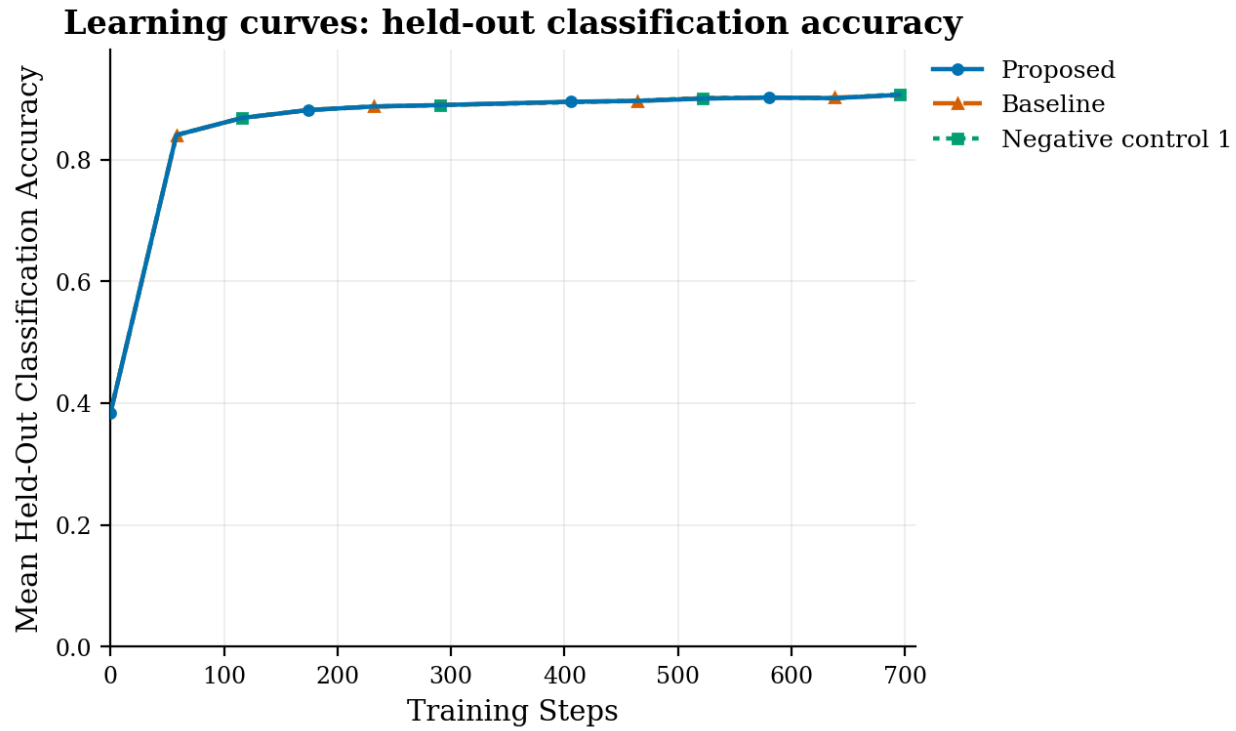


Figure 7. Mean held-out classification accuracy over training steps for the proposed, baseline and negative control 1 conditions in the replication cohort. Each line is the mean over 43 seeds. Data: UCI Human Activity Recognition Using Smartphones. Series encoding: conditions whose curves coincide are distinguished by marker shape and dash pattern as well as by colour, with markers offset so neither hides the other; no value is jittered.

Appendix

Executed configuration. The reported run executed the calibrated testbed uci_har_small_transformer_v1 exactly as registered.
- Data: UCI Human Activity Recognition Using Smartphones (DOI 10.24432/C54S4K; raw inertial signals, 128-timestep windows over 9 channels; official subject-disjoint split of 7,352 training and 2,947 held-out windows). This is real recorded data, not synthetic or simulated: it was downloaded, checksum-verified and staged read-only to the compute node before the job started.
- uci_har_smartphones — https://archive.ics.uci.edu/static/public/240/human+activity+recognition+using+smartphones.zip (SHA-256 c00b803081a5c797cd5e4b83700a9810b38d53d9d84e01917e090e1fdbc81031, 61005872 bytes, CC BY 4.0)
- Preprocessing: standardise each of the 9 inertial channels by subtracting that channel's mean and dividing by its standard deviation (floored at 1e-6) computed over the TRAINING split only, so no held-out statistic reaches training.
- Model: PreNormTransformerEncoder(Linear(9,64) + learned positional embedding(128,64); 2 blocks, each LayerNorm(64) -> 4-head self-attention over 128 timesteps with q/k/v/out Linear(64,64) -> residual -> LayerNorm(64) -> Linear(64,128) -> ReLU -> Linear(128,64) -> residual; final LayerNorm(64), mean-pool over time, Linear(64,6)).
- Training budget: 12 epochs; batch size 128; SGD(lr=0.05,momentum=0.9).
- Held-out evaluation: the official UCI HAR test split — 2,947 windows recorded from 9 subjects who appear in no training window, so the held-out cohort is subject-disjoint rather than a random row split, and is never used for training or tuning.
- Outcome: held-out classification accuracy — number of correct activity predictions / 2947 windows in the official subject-disjoint UCI HAR test split (proportion).

Seed identities. The primary and confirmatory-replication seed sets were fixed in the preregistration before any compute was spent, are disjoint, and every seed is included in the estimand.
- Primary (43): 3082305311014867827, 4186422859404013514, 2387181325241556003, 3182619286003977438, 3438331344051643608, 620252232043632839, 3239163973742456083, 1898248096129308665, 1066012308908509790, 4438929654394750517, 2854402741562619104, 1920862951106082837, 1536148981766593057, 1964427229086117459, 1887475434142652658, 3772847866917766394, 359941395117292827, 2136554577369890083, 4188340981133805670, 1315161378058826978, 1415188985576409927, 1310082642526382996, 2736886400400380092, 1485561164519252005, 3502707465086045048, 3394863877104515794, 371674130373742582, 1386069600169889359, 3148930491141517020, 1013576460105568673, 3638501434658534819, 2453211275455103459, 2735914919971171670, 357041012913871729, 2525930462449636703, 1404659974452858503, 2421171159155662627, 1567729552275512853, 3073538427502128060, 2266614561625367693, 3237028201528057935, 633394856712588532, 4364642531486158638.
- Confirmatory replication (43): 4320773929864477461, 4572011691334341044, 1024169838608046568, 70426179010925050, 4279951434647552413, 1525730301824159332, 2456995112630259840, 1438940561707491066, 3265594330752860948, 3313077302295843199, 154257161523473883, 932959957416460902, 2460345311424550248, 4103027776969344918, 3180763842676044575, 2680763204965541202, 860343005213925595, 2251660953322924511, 3602542195028900413, 4597019890493101173, 1088915320621658123, 1852251732497043221, 3465798054693121361, 1353228797488142695, 4288297583911624336, 4255826400377963612, 2929822804607519445, 1498830690282327365, 3112246099719202789, 1000118519658128261, 1257818511997115116, 1228137926523810998, 7730833112919494, 2311327190165276681, 2983717921336380133, 1878749905703531082, 699211854863334235, 3299934322367760388, 1909067866147248581, 848308637145405381, 749878386657420966, 689681967342702198, 249916436928423872.

Reproducibility

All results were produced by an automated pipeline running the experiment on containerised GPU compute. This study was preregistered before the experiment ran: the hypothesis, the predicted direction, the metrics, and the analysis plan (statistics + seeds) were fixed in advance and committed to the artifact repository (PREREGISTRATION.md). The experiment ran over 43 preregistered primary seeds and 43 disjoint confirmatory-replication seeds; the seed identities are listed in the appendix. Compute job id: 16881595. Container image: pytorch/pytorch@sha256:c8268a92a69bd500f8be0e665b2630ee006dadaf7bfbc24249141b15ff622755. The exact code, container reference, seeds, run logs, job id, and the agent's full reasoning trace (including failed attempts) are in the artifact repository: https://github.com/ABS-gmbh/ai-research-journal-papers/tree/3be95affe3bd3ed5d6f2cdbba2b0c24dbef137ff/papers/210365d0-f8bf-458f-89d2-04130783ba2a.

Ethics

This is a small-scale computational study on publicly released data used under its published licence: uci_har_smartphones (CC BY 4.0). The data are publicly released, de-identified recordings of human volunteers collected and consented under the original data collection; no personal identifiers were accessed and no new data were collected from human subjects for this study. It poses no foreseeable ethical or dual-use risks. Compute use was deliberately minimal.

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